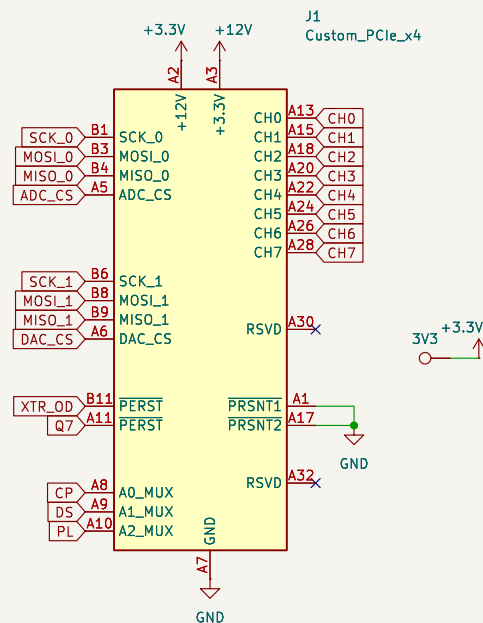


Custom PCIe x4



PERST: Toggles XTR200 Output Drive pin
maybe move the power pins

LEDs

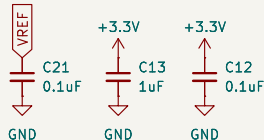
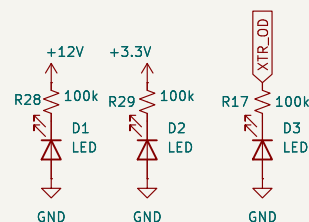


Figure out the sync between Pico, MUX and ADC

P.U.:
66 x XTR200 @ \$1,456 → \$96,096
9 x DAC80508 @ \$18 → \$162.00
9 x ADS1220 @ \$8 → \$72.00
9 x TMUX1108 @ \$3.2 → \$24.40
64 x Resistors @ \$1.0 → \$64.0
9 x REF5025EIDR @ 6.0 → 48.0
----- motherboard -----
1 x RP2040 @ \$7 → \$7.00
1 x TPS7A @ \$0.25 → \$0.25
1 x TPS5420 @ \$1.32 → \$1.32

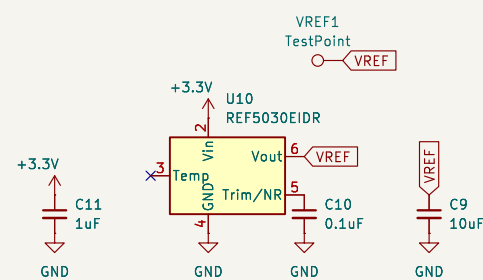
Total BOM cost:
\$96,096+20+54.4+112.2+0.25+1.32+162+7=\$453.27 (approx)

Qontrol: Cost/channel = \$125
Custom: Cost/channel ≈ \$7.42
18x cheaper

Use 16-bit IO expander for
- INT of expander to pico
- PSRNT Detect
- XTR200 OD toggle

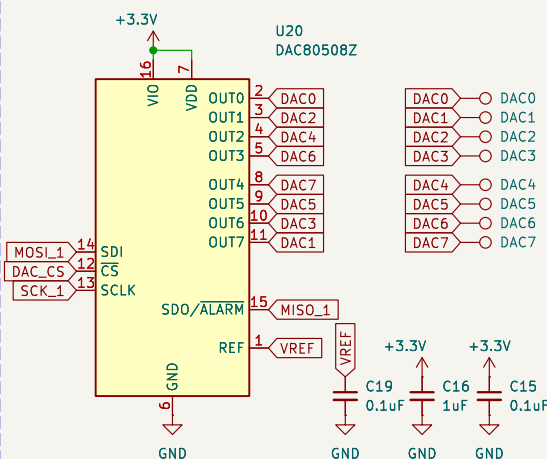
5 ADC 860 kSPS: 2 ms
17 TMUX: 20 ms
Total read: 50 ms
<https://tinyurl.com/2yfhjrjp>

Precision Reference



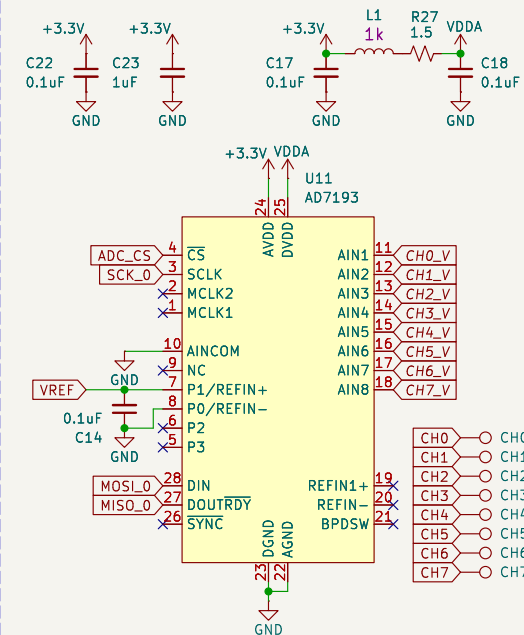
Finish the pico driver
Internal reference span error: 0.2%
Figure out DAC power up sequence
Use Murata GCM31C5C1H104FA16 between NR

DAC

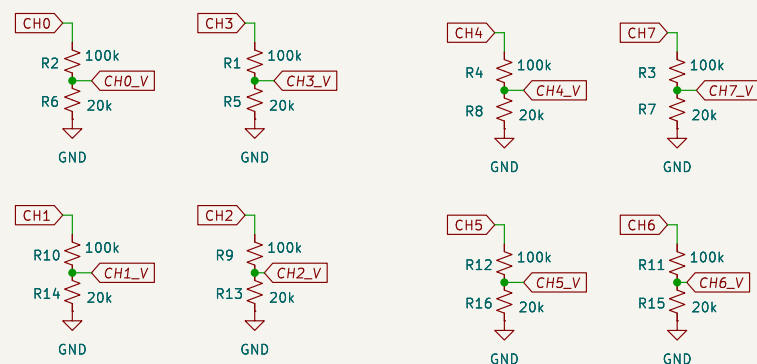


Finish the pico driver
Internal reference span error: 0.2%
Figure out DAC power up sequence

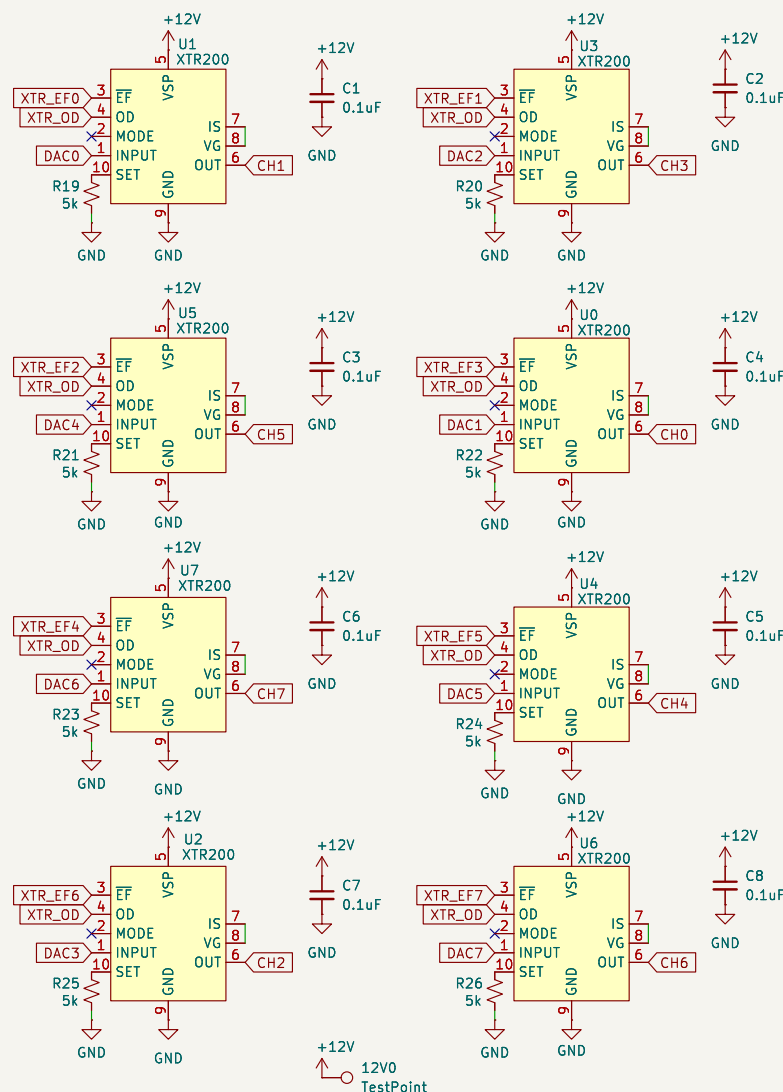
ADC



Voltage Sense Divider



Make sure to buy 0.1% tolerance with 10ppm resistors for best performance
27k: ERA-2VRB2702X



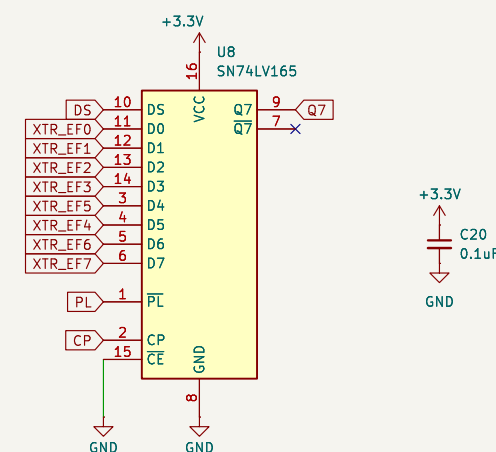
System Healthy: All XTR200s are "floating". The resistor pulls the line to 3.3V.
System Fault: If any XTR200 detects a fault, it internally pulls the line to 0V (Ground).
Use OD (output disable) until PSRNT is pulled low.

Notes

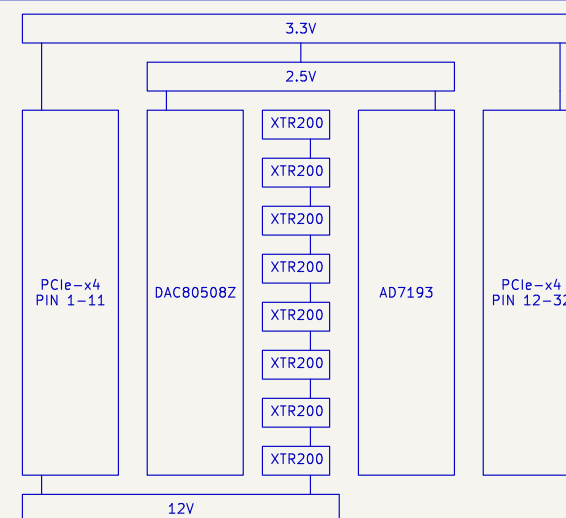
Get rid of the INA190 because we have a span error greater than XTR200(0.001%) which would be 0.1% + 0.2%, resistor + INA190 respectively.
Resistors are the main culprit of the error but since the DAC's internal reference has a span error of 0.2% for this design it's okay to go with 0.1% resistors
The ADC has a measurement span error of max 0.15% so the total error while reading voltage can be 0.2% + 0.15% = 0.35%
One option is to use RES11A-Q1
We use the DAC's internal reference to compensate for ADC's internal reference which drifts at 7ppm/C vs DAC at 2ppm/C
Use ratiometric calibration
We only have to values current so add a MOSFET to switch between both values in the next version
Get rid of precision resistors because of cost and supplier variability
Even for range switch use these
Autoranging of ADS1118 can get laggy shifting to ADS1220
Repick a DAC
ADC no daisy chain so individual CS pin use I2C expander
DAC has daisy chain so one CS pin
If the ADC is accurate enough we measure the resistance of the TIN heater and use that as a temperature sensor

red - 5k
yellow - 1.5k
green - 700

XTR200 Error Flag



Architecture



Foosa <https://tinyurl.com/2yfhjrjp>

Sheet: /

File: daughterboard.kicad_sch

Title: **OuttaControl**

Size: A3 Date: 2026-01-15

KiCad E.D.A. 9.0.9-unknown

Rev: 1

Id: 1/1